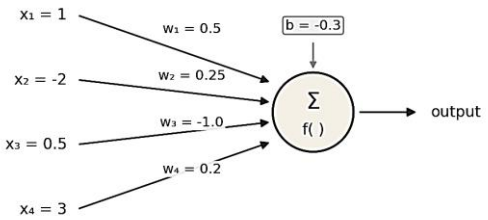
UNIT 1

1. Explain the structure of a Multilayer Perceptron . Describe the role of weights, biases, hidden layers, and activation functions in learning a nonlinear relationship.
2. Explain the mathematical model of an artificial neuron. Describe the role of weighted summation, bias, and activation function.
3. Explain why nonlinear activation functions are necessary in a deep neural network. What would happen if every layer used only a linear activation function?
4. Distinguish Sigmoid, Tanh, and ReLU activation functions with respect to output range, advantages, limitations, and suitable use in deep neural networks.
5. Explain the vanishing gradient problem in deep neural networks. Discuss how activation functions and increasing network depth can contribute to this problem.
6. Explain the process of backpropagation in a feedforward neural network. Illustrate and derive the flow of input data, predicted output, loss, and gradients using a neat diagram.
7. Consider a neural network having two input neurons, two hidden neurons, and one output neuron. The inputs are $x_1 = 1.0$ and $x_2 = 2.0$. The input-to-hidden weights are $w_{11} = 0.2$, $w_{12} = 0.4$, $w_{21} = 0.3$, and $w_{22} = -0.2$, where w_{ij} connects input x_i to hidden neuron h_j. Hidden biases are $b_{h1} = 0.1$ and $b_{h2} = 0.2$. Use ReLU activation. The hidden-to-output weights are $v_1 = 0.5$ and $v_2 = -0.4$, and output bias $b_o = 0.1$. Use sigmoid activation. (i) Illustrate the complete neural network architecture clearly showing weights and biases. (ii) Evaluate the outputs of h_1 and h_2. (iii) Evaluate the final network output. Use $\sigma(z) = \frac{1}{1+e^{(-z)}}$
8. For a binary classification problem, a neural network produces the predicted output: $\hat{y} = 0.72$ for a training sample whose actual output is: $y = 1$The loss function is defined as:

$$L = ½(y - \hat{y})^2$$

Evaluate the following:
(i) The value of the loss, L.
(ii) The gradient of the loss with respect to the predicted output: $\partial L/\partial \hat{y}$
(iii) If the output neuron uses the sigmoid activation function, calculate the gradient of the loss with respect to the neuron's pre-activation value, z, using: $\partial L/\partial z = (\partial L/\partial \hat{y}) \cdot \hat{y}(1 - \hat{y})$

9. A single artificial neuron receives four inputs with the weights and bias shown in the figure.



(i) Determine the net input (pre-activation) z of the neuron.

(ii) Evaluate the neuron output if f is the sigmoid function.

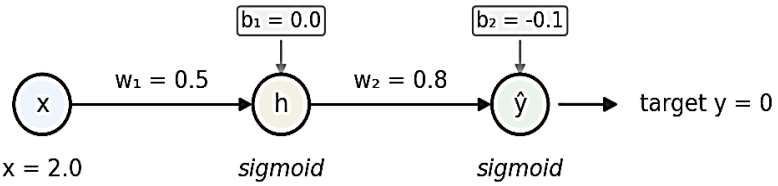
(iii) Estimate the neuron output if f is tanh, if f is ReLU, and if f is Leaky ReLU with a negative slope of 0.01.

(iv) Explain which of these activations would stop the gradient from flowing back through this neuron, and why.

10. Consider a two-layer neural network with 2 input neurons, 3 hidden neurons, and 1 output neuron. All connections are fully connected.

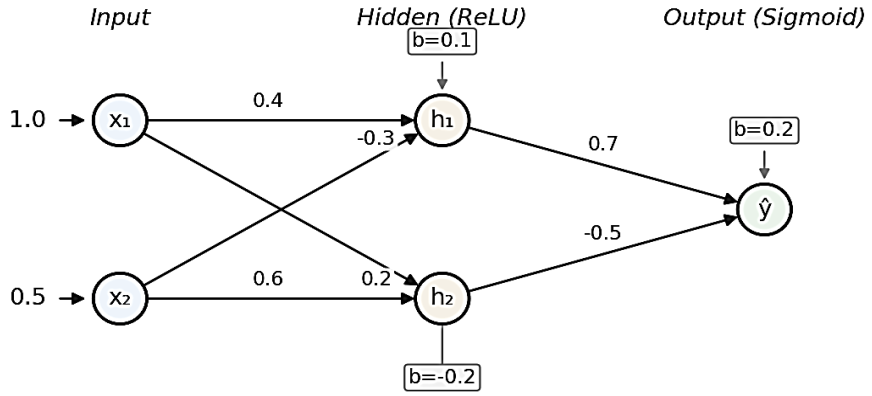
(i) Draw the architecture.
(ii) Evaluate total number of weights.
(iii) Evaluate total trainable parameters when each hidden and output neuron has one bias.
(iv) Explain how the parameter count changes if hidden neurons increase from 3 to 5.

11. In the network below both neurons use the sigmoid activation. The loss is $L = \frac{1}{2}(y - \hat{y})^2$ and the learning rate is $\eta = 0.1$.



(v)
(vi) (i) Perform the forward pass: calculate z_1, h, z_2 and $\hat{y}$, and hence the loss L.
(vii) (ii) Calculate $\partial L/\partial \hat{y}$ and the output-layer error term
(viii) $\delta_o = (\partial L/\partial \hat{y}) \cdot \hat{y}(1 - \hat{y})$.
(ix) (iii) Calculate $\partial L/\partial w_2$ and $\partial L/\partial b_2$.
(x) (iv) Calculate the hidden-layer error term δ_h and hence $\partial L/\partial w_1$ and $\partial L/\partial b_1$.
(xi) (v) Update w_1 and w_2 using gradient descent, and comment on why the update to w_1 is so much smaller than the update to w_2.

12. The network below is fully specified. All hidden neurons use ReLU and the output neuron uses the sigmoid function.



(i) Compute the pre-activation values z_{h1} and z_{h2}, and hence the hidden-layer outputs h_1 and h_2.

(ii) Compute the pre-activation value at the output neuron and the final network output $\hat{y}$.

(iii) If the target output is $y = 1$ and the loss is $L = \frac{1}{2}(y - \hat{y})^2$, calculate the loss.

(iv) State what the value of h_1 would become if the bias $b = 0.1$ were changed to -0.5, and explain what this reveals about ReLU.